David Yuanze Li

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Student researcher interested in applied mathematics, cybersecurity, and AI methods for human health.

EDUCATION

2023—present **Thousand Oaks High School, Thousand Oaks, CA**

- Class of 2027; Center for Advanced Studies and Research
- Weighted GPA: 4.77 • Unweighted GPA: 3.98
- Athlete of the cross-country team

2022—2026 **Moorpark College, Moorpark, CA**

- Introduction to Statistics (A) • An Introduction to Astronomy (A) • Programming Fundamentals (A)

2026 **Coastline College, Fountain Valley, CA**

- Introduction to Cybersecurity (A) • Network Security (Security+) (A)

COMPETITIONS AND AWARDS

2026 **Late-breaking abstract and poster, 48th Annual International Conference of IEEE Engineering in Medicine and Biology Society Conference (IEEE EMBC 2026), Toronto, Canada**

- "*Towards Natural Speech: Attention-Based Detection, Localization, and Reference-Free Fluency Reconstruction of Stuttering Speech*." Presented July 29, 2026.

2026 **Poster Presentation, 117th Annual Conference of the American Association for Cancer Research (AACR 2026), San Diego, California**

- "*Plasma Cell-Free DNA Methylation Markers for Cancer Detection*." Presented at the Program: The Conquest of Cancer and the Next Generation of Cancer Researchers.

2026 **Qualifier, California State Science and Engineering Fair**

- Project: *AI-based stuttering detection and fluent-speech reconstruction project*.

2025—2026 **1st place & 2nd place, Ventura County Science Fair**

- Project: *AI-based stuttering detection and fluent-speech reconstruction project*.

2026 **Qualifier, American Invitational Mathematics Examination (AIME)**

- Qualified in 11th on the AMC 12 (Top ~5% nationally).

2025 **Oral presentation and travel award, 13th International Conference on Intelligent Biology and Medicine (ICIBM 2025), Columbus, Ohio**

- "*AI-based stuttering detection*." Achieved the travel award.

2022—2025 **CyberPatriot semifinal; multiple times of Platinum tier in state round**

- The Youth Cyber Defense Competition by Air Force Association, the world's largest cybersecurity competition, open to US & Canada schools.

2024 **Bronze medal, Math League International Tournament**

2023—2024 **President's Volunteer Service Award (Silver + Bronze)**

PUBLICATIONS

2026 **Toward the Simultaneous Detection of Multiple Diseases with a Highly Cost-Effective Cell-Free DNA Methylome Test**

- Authors: Weihua Zeng, ..., **David Li**, ..., Jasmine Zhou
- Journal: ***Proceedings of the National Academy of Sciences of the United States of America (PNAS)***
- Status: Published. Web link: https://doi.org/10.1073/pnas.2518347123
- News highlight by NBC News and U.S. News & World Report. Check media coverage summary.

2026 **When RSA Encryption Becomes Vulnerable: The Role of Prime Separation in Fermat Factorization**

- Authors: **David Li**
- Status: Completed and will be submitted. Web link: https://dliport.github.io/assets/math.cybersecurity.pdf

2026 **A Privacy-Preserving On-Device AI System for Stuttering Detection and Reference-Free Fluency Reconstruction on Smartphones**

- Authors: **David Li** and Min Xu
- Journal: ***Journal of Fluency Disorders***
- Status: Submitted. Web link: https://dliport.github.io/assets/ai.stuttering.privacy.preserving.pdf

2026 **Plasma Cell-Free DNA Methylation Markers for Cancer Detection**

- Authors: **David Li** and Frank Alber
- Conference poster: ***117th Annual Conference of the American Association for Cancer Research (AACR 2026), San Diego, California***
- Status: Accepted

2026 **Towards Natural Speech: Attention-Based Detection, Localization, and Reference-Free Fluency Reconstruction of Stuttering Speech**

- Authors: **David Li** and Min Xu
- Conference abstract: ***48th Annual International Conference of IEEE Engineering in Medicine and Biology Society Conference (IEEE EMBC 2026), Toronto, Canada***
- Status: Accepted

2025 **Empowering Confident Communication: Artificial Intelligence Based Detection of Stuttering Patterns for Speech Therapy**

- Authors: **David Li** and Frank Alber
- Conference poster: ***13th International Conference on Intelligent Biology and Medicine (ICIBM 2025), Columbus, Ohio***
- Status: Presented and received travel award

SOFTWARE AND APP

2026 **SPEFA: Speech Fluency: Apple iOS app and Android app**



- Authors: **David Li**
- Download link: https://dliport.github.io/spefa/

ATHLETE EXPERIENCES AND AWARDS

2023—2025 **Cross-country / Track-and-field runner, Thousand Oaks High School**

2024 **3rd place, Ventura County two-mile cross-country race**

2024 **Qualifier, Tiger Invitational Rising Stars 800-meter meet**

RESEARCH/INTERN EXPERIENCES

2025—present **Research intern, Carnegie Mellon University**

- Mentor: Professor Min Xu
- Developed AI methods and smartphone apps to detect stuttering and reconstruct fluent speech.

2024—present **Volunteer researcher, UCLA**

- Mentor: Professor Frank Alber
- Analyzed cell-free DNA methylation markers for a blood test of simultaneously detecting multiple diseases.

2025 **Intern for programming & AI, EarlyDiagnostics Inc.**

- Programming for data processing pipelines, AI foundation model, and database maintenance.

LEADERSHIP EXPERIENCE

2025—present **Instructor, EmpowerNext Research Literacy Program**

- Guide middle school students through reading and discussing research literature.

2024—present **President, Cyber Watch, Thousand Oaks High School**

- Lead club meetings and teach cybersecurity concepts while preparing members for competitions.

2025 **CyberPatriot team coach, Thousand Oaks High School**

- Teach cybersecurity skills and mentor teammates in cybersecurity problem solving.
- The team that I coached achieved ***semifinal of Cyberpatriot 2025 at the Gold Tier***.

2022—present **Head of Youth Altar Servers, St. Jude the Apostle Church**

- Train new altar servers, coordinate duties and schedules, and keep the youth team informed.

2023—2026 **Multimedia content creator**

- Produce videos about the online game taming.io; approximately 660 subscribers as of September 2026. Channel: youtube.com/@DragonFire2271

VOLUNTEER EXPERIENCE

2025—present **Math tutor, Thousand Oaks High School**

- Tutor high school students in mathematics.

2023 **Community food bank, Thousand Oaks, CA**

- Distributed food to more than 1,000 people.

2023—2024 **Rose Parade volunteer, Fontana, CA**

- Prepared floral materials, painted, decorated, and cleaned floats.

2021—present **Youth altar server, St. Jude the Apostle Church**

- Serve at weekend Masses.

2022—present **STEM tutor, Guadalupe Community Services Center, Canoga Park, CA**

- Teach elementary school students math and science.

2022—present **Volunteer tutor, Thousand Oaks High School help center**

- Provide regular math help for students.

2022—present **Food collection volunteer, St. Jude the Apostle Church**

TRAINING

2026 **UC Davis COSMOS, Applied Mathematics**

- Advisor: Professor Janko Gravner
- Studied set theory, topology, probability, and encryption algorithms; researched and presented "***The Foundations of RSA Encryption and the Vulnerabilities of Proximity: An Analysis of Fermat's Factorization***."

2025 **Machine Learning Boot Camp**

- Studied machine learning and AI methods.

SKILLS

- Fluent in English and Chinese; basic Spanish.
- Python, Bash scripting, Linux and Windows operating systems, machine learning, cybersecurity, and networking techniques.